

DATA SCIENTIST INTERN ASSIGNMENT

Technical Report

Driver Behaviour and Vehicle Health Dashboards from One Week of Two-Wheeler Fleet Telemetry

Candidate organisation	VexarDrive Technologies
Dataset	VEXAR_Fleet_Dataset_CANDIDATE_VERSION.xlsx
Observation window	31 July 2026 - 6 August 2026
Scope	30 drivers, 30 two-wheelers, 450 trips, 12,987 per-minute GPS + IMU rows
Geography	Bengaluru hubs (Rajajinagar, HSR Layout, Bellandur, and others)
Code and dashboards	https://github.com/nishakumari26/VexarDrive-Fleet-Intern-Assignment
Interactive HTML	dashboards/driver_behaviour.html and dashboards/vehicle_health.html

This report documents how every KPI, threshold, score and band on the two required dashboards was derived from the candidate workbook. No synthetic rows were added. Cut-offs are empirical percentiles of this week's moving telemetry, not textbook automotive constants or legal speed limits.

1. Problem and required outputs

VexarDrive operates a two-wheeler delivery fleet. The candidate brief asks for two explainable dashboards built only from the supplied week of trip and phone-sensor data:

- 1 Driver Behaviour Dashboard** - score every driver, separate risky vs safe patterns, and show why each score exists.
- 2 Vehicle Health Status Dashboard** - flag vehicles whose IMU / GPS / trip patterns suggest wear or a workshop visit, and show what triggered the flag.

The brief also requires a methodology for every displayed number, explicit assumptions, and additional uses of the same columns. Labels such as Safe / Risky or Healthy / Maintenance Required are used only where the data can support a relative ranking of the current 30-unit roster.

2. Data loading, joins and validation

Sheets are read with **header=2** because row 0 is a title (for example "Drivers (Master Data)"), row 1 is blank, and row 2 holds column names. Joins follow the brief:

Telemetry.Trip_ID → Trips.Trip_ID | Trips.Driver_ID → Drivers.Driver_ID | Trips.Vehicle_ID → Vehicles.Vehicle_ID

Quality checks written to outputs/qa_report.json:

Check	Result
Rows	30 drivers, 30 vehicles, 450 trips, 12,987 telemetry minutes
Null cells	0 on all four tables
Orphan keys	0 (every trip driver/vehicle exists in master data)
Telemetry coverage	Every trip has telemetry; row count = Duration_Min
ID consistency	Telemetry Driver_ID and Vehicle_ID match the parent trip
Balance	Exactly 15 trips per driver

Two data-quality notes were recorded and then excluded from scoring: two people share the name Kavya Pillai (D07 and D23), so analysis keys on Driver_ID; Gender sometimes disagrees with the given name, so gender is not an input to any KPI.

3. Why thresholds are empirical

The phone IMU is one sample per minute. A literature harsh-brake cut-off of about 0.4 g lasting about one second cannot be applied honestly: a 60-second average hides the spike. Accel_Z has mean 1.007 g (gravity on Z). One-minute change in speed is uncorrelated with the accel axes, so speed delta is not used as a harsh-event detector.

Event lines are the 95th percentile of **moving** minutes (speed at least 5 km/h; n = 11,891). Five km/h is only an idle/crawl gate, not a safety limit.

Signal	Event line (P95)	Notes
Speed_kmph	42.4 km/h	P99 = 52.1 km/h. Not a posted legal limit (none in the file).
Horizontal accel $\sqrt{X^2+Y^2}$	0.460 g	Phone-plane jerks; Z is gravity-dominated.
Gyro_Z	5.01 deg/s	Heavy tail (P99 about 39.7 deg/s).
Accel_Z - 1 g	0.087 g	Vertical bump / suspension proxy.

Counts are converted to rates per 100 km using Trips.Distance_KM so a longer trip is not punished only for lasting more minutes.

4. Driver Behaviour Dashboard

4.1 Metrics

KPI	Columns	Calculation	Why it is relevant
Overspeed min / 100 km	Speed_kmph, Distance_KM	Minutes with speed > 42.4 km/h, divided by km, times 100; mean of 15 trips.	Only direct collision-energy / speed signal in the workbook.
Harsh horizontal IMU / 100 km	Accel_X_g, Accel_Y_g	$\sqrt{X^2+Y^2} > 0.460$ g, exposure-adjusted.	Proxy for aggressive start/stop or swerve when 1-minute speed deltas are too smooth.
Yaw events / 100 km	Gyro_Z_dps	$ \text{Gyro_Z} > 5.01$ deg/s, exposure-adjusted.	Abrupt heading change on a two-wheeler.
Mean trip max speed	Max_Speed_kmph	Average of trip maxima.	Captures a short peak that may not dominate the overspeed-minute share.

4.2 Overall risk score

Each metric is converted to a percentile rank among the 30 drivers (0-100). The composite is:

$$\text{risk_score} = 0.30 \times \text{overspeed rank} + 0.25 \times \text{harsh IMU rank} + 0.25 \times \text{yaw rank} + 0.20 \times \text{max-speed rank}$$

Ranks avoid arbitrary 0-1 scaling. Weights put slightly more mass on overspeed because it is the most interpretable safety signal here; IMU components are equal. Changing the weights reorders the middle of the pack more than the extremes. This weighting is an assumption and is stated as such.

4.3 Bands (Safe / Moderate / Risky)

There are no labelled crashes or insurance events. With $n = 30$, the only classification justified by the data is a relative split of this week's scores. Tertiles give a 10 / 10 / 10 split:

Band	Rule on risk_score	Meaning
Safe	score ≤ 37.0	Safer than peers this week, not 'legal'.
Moderate	$37.0 < \text{score} \leq 64.3$	Typical for this roster.
Risky	score > 64.3	Riskier than peers this week, not 'illegal'.

The dashboard table includes a full 'why' sentence: the four rates, kilometres, trips, and the largest contributing rank, so a reviewer can see why a driver received a particular score.

4.4 Findings this week

ID	Driver	Risk	Band	Largest contributor	Overspeed /100km
D19	Senthil Pillai	95.0	Risky	overspeed minutes / 100 km	53.1
D14	Rajesh Subramaniam	93.7	Risky	harsh horizontal IMU spikes / 100 km	38.5
D06	Bhavani Raj	93.0	Risky	high yaw-rate minutes / 100 km	33.8
D03	Vignesh Murugan	89.8	Risky	harsh horizontal IMU spikes / 100 km	24.1
D24	Lakshmi Iyer	87.5	Risky	overspeed minutes / 100 km	34.0

Table 1. Highest-risk drivers this week.

ID	Driver	Risk	Band	Largest contributor
D28	Praveen Iyer	9.5	Safe	high yaw-rate minutes / 100 km
D09	Rajesh Raj	11.2	Safe	mean trip max speed
D11	Mohan Iyer	12.2	Safe	high yaw-rate minutes / 100 km
D22	Priya Iyer	15.5	Safe	overspeed minutes / 100 km
D05	Manoj Kumar	17.7	Safe	overspeed minutes / 100 km

Table 2. Lowest-risk (safest vs peers) drivers this week.

Headline: D19 Senthil Pillai scores 95.0 (Risky) because overspeed is 53.1 minutes per 100 km (fleet-worst). D28 is at the other end of the same relative scale (9.5, Safe).

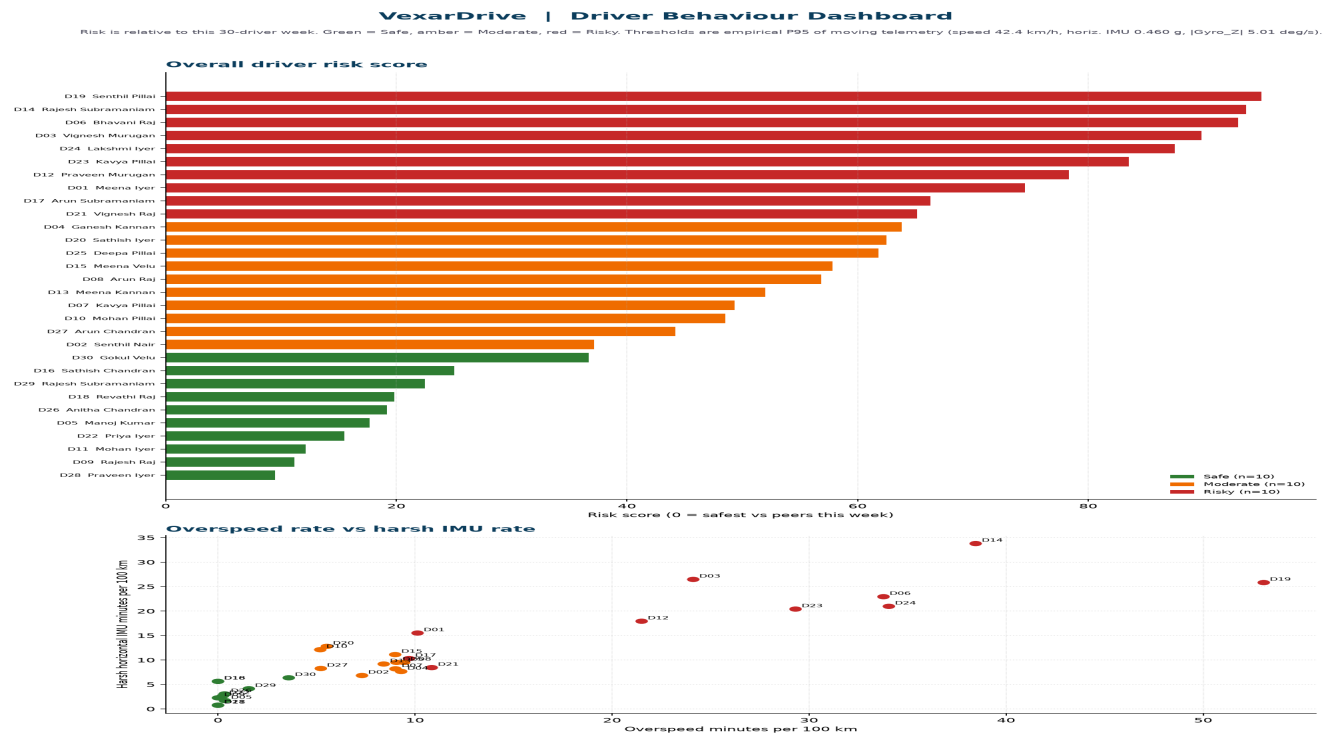


Figure 1. Driver Behaviour Dashboard (static image of the interactive HTML).

5. Vehicle Health Status Dashboard

The workbook has no OBD, DTCs, tyre pressure or workshop job cards. Health is inferred from master-data wear plus speed-controlled IMU. GPS is used as Speed_kmph (dictionary: instantaneous speed). Latitude / longitude are not in the health score because there is no planned route or road-quality layer; they remain a pothole-heatmap extra use case.

Cruise window 15-35 km/h matches this fleet's typical urban speed (trip average about 24 km/h) and reduces stop-go riding style so vibration is closer to a machine signal.

5.1 Metrics

KPI	Columns	Calculation	Why it is relevant
Days since service	Last_Service_Date vs 6 Aug 2026	Days late, then fleet percentile rank.	No OEM interval in the file; lateness is relative.
Odometer	Odometer_KM_Start_of_Week	Percentile rank.	Only mileage field; wear accumulates with distance.
Age (years)	Manufacture_Year	2026 minus year, then rank.	Older two-wheelers typically need more shop time.
Cruise vibration	Accel X/Y/Z at 15-35 km/h	Mean accel magnitude - 1 g , then rank.	Persistent shake at steady speed (engine / wheel / suspension).
Cruise gyro	Gyro X/Y/Z at 15-35 km/h	Median gyro magnitude, then rank.	Median used because gyro has a heavy tail; rotational shake.
Vertical bumps / 100 km	Accel_Z_g, Distance_KM	Z-1 > 0.087 g, exposure-adjusted.	Pothole or worn-suspension proxy.

5.2 Health / maintenance score

$$\text{maintenance_risk} = 0.22 \text{ service} + 0.18 \text{ odometer} + 0.12 \text{ age} + 0.22 \text{ vibration} + 0.12 \text{ cruise gyro} + 0.14 \text{ bumps}$$
$$\text{health_score} = 100 - \text{maintenance_risk}$$

Workshop fields (service, odometer, age) do not depend on riding style. IMU terms are speed-controlled. Weights are assumptions: slightly more mass on service recency and cruise vibration.

5.3 Bands

Band	Rule on maintenance_risk	Meaning
Healthy	risk <= 36.0	Healthier than peers this week.
Needs Attention	36.0 < risk <= 65.4	Typical for this roster.
Maintenance Required	risk > 65.4	Priority workshop queue vs peers, not a DTC.

5.4 Findings and confounding

ID	Vehicle	Health	Band	Odo km	Days since service	Largest contributor
V02	TVS Raider	4.8	Maintenance Required	46,601	92	days since last service
V12	TVS Ntorq	17.1	Maintenance Required	31,966	58	odometer at week start
V01	Yamaha Ray ZR	17.7	Maintenance Required	22,820	59	days since last service
V23	Suzuki Access	19.7	Maintenance Required	17,821	51	cruise accel magnitude - 1 g
V10	TVS Ntorq	19.8	Maintenance Required	30,162	36	median cruise gyro magnitude

Table 3. Highest maintenance-risk vehicles (shop first).

Priority unit: V02 TVS Raider, health 4.8, 46,601 km, 92 days since service. Nineteen vehicles have a single rider this week, so IMU can still mix style with machine. Mitigation: (1) service / odometer / age are rider-independent; (2) vibration is cruise-only; (3) V02's usual rider D02 is only Moderate on behaviour while V02 is the worst health unit - evidence that the vehicle metrics are not a copy of the driver score.

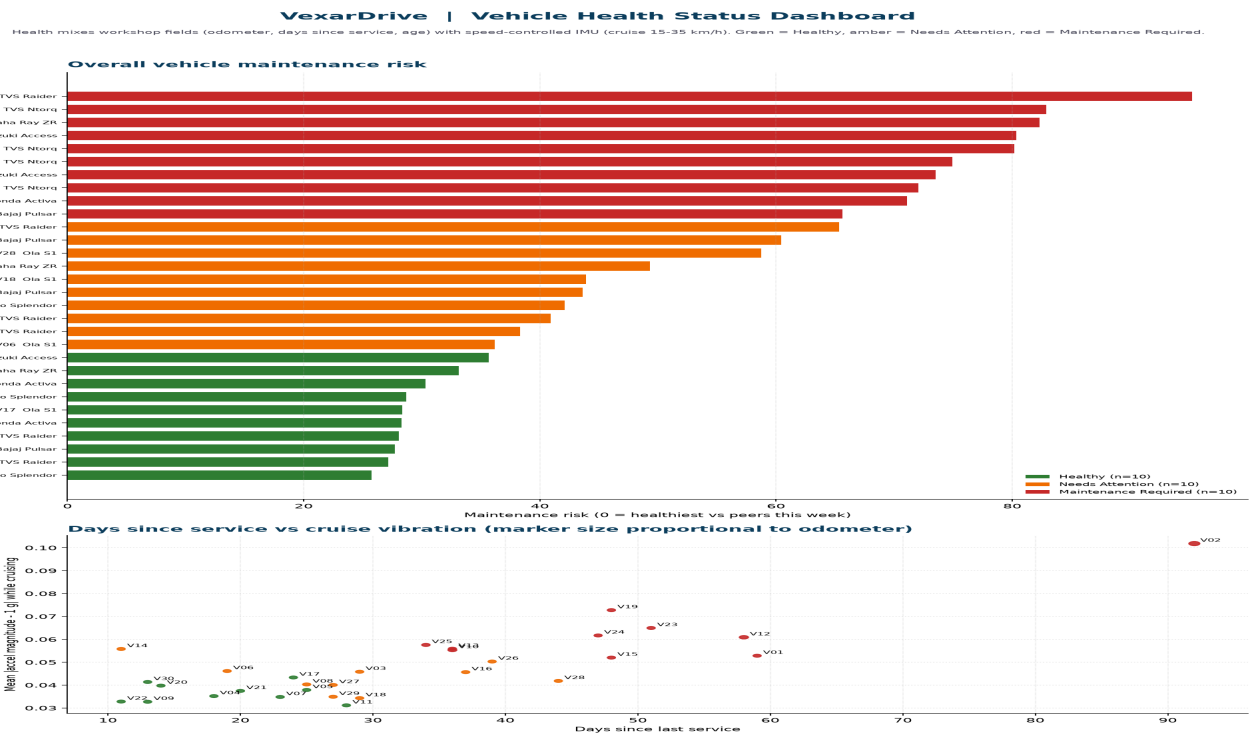


Figure 2. Vehicle Health Status Dashboard (static image of the interactive HTML).

6. Assumptions

- 1 One week is enough to **rank** the current roster, not to predict crashes.

- 2 Phone orientation is stable enough that Z is approximately gravity (supported by mean Z about 1.0 g).
- 3 P95 event lines mean 'unusual for this fleet', not physics constants.
- 4 Tertile bands are the only classification justified by $n = 30$ and the absence of labelled accidents.
- 5 Trips.Distance_KM is the exposure base (data dictionary: derived from per-minute speed).
- 6 Score weights are stated assumptions, not estimated from a labelled loss function.

7. What this dataset cannot support

The following were not invented: fuel use, idling cost, engine temperature, brake-pad wear, crash probability, legal speeding against a posted limit (no speed-limit layer), or a 'safe' g-threshold copied from automotive literature. If a metric cannot be computed from the columns, it is omitted rather than fabricated.

8. Additional use cases supported by the same columns

- 1 **Hub workload** - trip counts and kilometres by Home_Hub / start coordinates.
- 2 **Time-of-day risk** - Start_Time plus overspeed minutes (evening vs morning).
- 3 **Rider-bike rotation** - 44 driver-vehicle pairs this week; V23 already has four riders, which helps isolate the bike.
- 4 **Service scheduling** - rank days-since-service times odometer without IMU.
- 5 **Route roughness map** - latitude/longitude of vertical-bump minutes (pothole heatmap).
- 6 **Onboarding quality** - License_Experience_Years vs risk rank (exploratory only; not causal).
- 7 **Coaching prototype** - the same event flags, if sampling later exceeds one sample per minute.

9. Reproducibility

From the repository root: `python -m pip install -r requirements.txt` then `python src/pipeline.py`. The script validates joins, derives P95 thresholds from this week's moving minutes, writes outputs/*.csv (including kpi_dictionary.csv with column / formula / why / threshold for every KPI) and rebuilds the HTML dashboards. Interactive dashboards: <https://github.com/nishakumari26/VexarDrive-Fleet-Intern-Assignment>.

Static dashboard images for reviewers who cannot open HTML are in submission/dashboard_images/. They show the same 30 drivers, 30 vehicles, scores and colour bands as the HTML files.

End of report. All numbers above are computed from VEXAR_Fleet_Dataset_CANDIDATE_VERSION.xlsx.